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SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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SDC HACKATHON 2026

Technical & Tools Declaration

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1. Team & Project Overview

Team Name	0x404
Project / Solution Name	GrantFlow
Name (Leader)	Khoo Zong Hao
Email & Phone (Leader)	zonghao04@gmail.com
University Affiliation(s)	Asia Pacific University
Live Prototype URL	https://sdchack.vercel.app/
Code Repository URL (public / access granted to judges)	https://github.com/JV-GG/SDCHACK.git

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

SME GrantFlow operates as an event-driven, decoupled system connecting five core layers end-to-end:
1. User Interface (Frontend): A responsive Next.js 16 (React 19, TypeScript, Tailwind CSS) web application hosted on Vercel serves the Grant Hub, Opportunity Radar, Review & Sign-Off modals, and Audit View.
2. Backend & Rules Engine: Client requests communicate via HTTPS/REST to a Python FastAPI service deployed on Replit. The backend executes deterministic Python rule modules for zero-hallucination quantitative evaluations (Malaysian equity, tenure, turnover caps, and approved TSP vendor matching).
3. AI Inference Pipeline: Unstructured invoice data and narrative drafting are routed through a resilient three-tier chain: primary DeepSeek Cloud API, secondary RunPod Serverless GPU (vLLM hosting deepseek-ai/DeepSeek-R1-Distill-Qwen-7B), and offline deterministic templates.
4. Data & Audit Layer: SME profiles, claims, and application packets are persisted in local SQLite with an automated Neon Serverless PostgreSQL mirror. Every event is written to an immutable, SHA-256 hash-chained audit trail.
5. Human Sign-Off & Dossier: Following owner sign-off, the engine compiles a standardized submission dossier sealed with a cryptographic SHA-256 fingerprint.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
Python 3.11+	Core backend language and execution runtime	Python Software Foundation (PSF) License
FastAPI (v0.115+)	High-performance asynchronous REST API framework handling event triggers, routing, and auth	MIT License
Pydantic (v2.9+) and Pydantic Settings	Strict data validation, contract modeling, and environment configuration management	MIT License
Uvicorn (v0.32+)	Production ASGI server implementation powering the FastAPI application	BSD 3-Clause License
HTTPX (v0.27+)	Asynchronous HTTP client for communicating with DeepSeek and RunPod AI endpoints	BSD 3-Clause License
Next.js (v16.4+) and React (v19.3+)	Modern frontend application framework and reactive component architecture	MIT License
TypeScript (v5+)	Static typing across the frontend codebase, synchronized with backend Pydantic models	Apache License 2.0
Tailwind CSS (v4+)	Utility-first CSS styling framework for responsive and accessible interface design	MIT License
Lucide React (v1.61+)	Open-source iconography for UI navigation, badges, and status indicators	ISC License
Pytest (v8.3+)	Automated test suite for validating deterministic grant rules, API endpoints, and audit chains	MIT License

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
DeepSeek Cloud API	Primary GenAI tier used for OCR parsing of unstructured invoice text, extracting line-item expenses, and drafting formal grant project justifications.	Paid/API per usage
RunPod Serverless vLLM API	Secondary GPU interface tier exposing an OpenAI-compatible /v1/chat/completions endpoint running deepseek-ai/DeepSeek-R1-Distill-Qwen-7B for resilient narrative generation and document parsing	Sponsor provided
Neon PostgreSQL Connection API	Serverless database connectivity (PostgreSQL wire protocol) used for cloud synchronization of SME profiles, grant applications, and hash-chained audit logs	Free tier

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit	Production hosting of the Python FastAPI backend service via .replit container configuration, secrets management, and continuous API availability for the client application.	Auto scale Deployment
RunPod	Serverless GPU compute infrastructure hosting a dedicated vLLM container worker with deepseek-ai/DeepSeek-R1-Distill-Qwen-7B model weights for heavy inference workloads.	Serverless GPU (endpoint reig5yleze9o5f)
Neon	Serverless cloud PostgreSQL database providing instant branching, storage, and automated failover mirroring for SME organization data and tamper-evident audit logs.	Serverless Free Tier
Vercel	Production edge hosting and deployment for the Next.js 16/React 19 frontend web application (sdchack.vercel.app), ensuring sub-second global asset delivery	Hobby

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
DeepSeek-V3 (deepseek-chat)	DeepSeek	Pre-trained (Zero-shot prompt engineering for unstructured invoice OCR parsing and narrative generation)	Commercial API Terms of Service
DeepSeek-R1-Distill-Qwen-7B	DeepSeek-AI/HuggingFace	Pre-trained (Open-weight reasoning model for fallback invoice item extraction and project justification drafting)	MIT License
Deterministic Template Engine	0x404 Team	Self-developed (Rule-based heuristic template drafter ensuring 100% offline availability)	MIT License

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Malaysian Government Grant Policy & Eligibility Rules Matrix	Compiled from public guidelines across official agency portals	Public	Open Government Information	N
MDEC Technology Service Provider (TSP) Directory	Official MDEC accredited vendor whitelist directory for MSME Digital Grant verification	Public	Public Information	N
Synthetic Malaysian SME Profiles & Invoices Dataset	Internal project repository	Synthetic	MIT License	N

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☒ Yes ☐ No

If yes, list them below:

Tool & Version	Used For	Extent of Use (minor / moderate / major)
Gemini 3.8 Flash	Next.js frontend component prototyping, responsive UI layout adjustment, utility classes, and quick iterative frontend bug fixes	Moderate
Claude Opus 5.5	High-level system architecture design, multi-agency grant data flow planning, deterministic rules engine architecture, and complex end-to-end integration strategy	Major
Claude Sonnet 5.5	Core backend service implementation FastAPI REST route handlers, Pydantic schema contract synchronization, and automated test suite authoring	Major

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

Every AI-assisted code snippet, schema, and architectural plan underwent strict human review before merging. We enforced strict end-to-end typing via TypeScript and Pydantic to eliminate runtime drift. To ensure absolute compliance with government grant criteria, all deterministic rules and endpoints were validated against a comprehensive 45-test automated pytest suite with zero tolerance for hallucinations. We manually verified all user journeys including invoice parsing, multi-agency grant evaluation, and cryptographic packet generation—to ensure full regulatory fidelity.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☐ Yes ☐ No

If yes, list them below:

Sponsor	Resource (API / credits / platform / data)	How It Was Used in the Project
RunPod	Serverless GPU Compute Platform and Credits	Deployed a dedicated serverless vLLM container endpoint running deepseek-ai/DeepSeek-R1-Distill-Qwen-7B for parsing unstructured invoice PDFs and drafting formal grant project justifications.
Replit	Cloud Development Environment and Autoscale Hosting Platform	Hosted the production FastAPI backend service via .replit container configuration, managed environment secrets, and provided public REST API access to the frontend.

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

The sponsor resources were essential in delivering a high-performance, production-grade cloud prototype. RunPod provided dedicated serverless GPU compute for private, low-latency LLM inference, while Replit enabled rapid deployment and autoscale hosting for our FastAPI backend. However, our architecture was intentionally designed with high resilience: if sponsor endpoints experience downtime, our system gracefully fails over across a 3-tier chain falling back from RunPod GPU to DeepSeek cloud API or built-in deterministic offline templates.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

GrantFlow collects minimal business data: account email, company profile parameters (SSM registration, sector, Malaysian equity %, headcount, turnover), and invoice line items. Personal NRICs, consumer data, and banking credentials are never collected.
Only itemized invoice descriptions and procurement categories are sent to AI endpoints (DeepSeek / RunPod) strictly for narrative drafting and company identifiers remain decoupled. All data in transit is encrypted using HTTPS/TLS 1.3. At rest, data is stored in isolated SQLite/Neon PostgreSQL databases, with all API credentials managed via server-side environment secrets. Every transaction appends to an immutable, SHA-256 hash-chained audit log recording event timestamps and IP addresses for tamper-evident tracking. Data is retained solely for the active session and application lifecycle. SME owners can update or remove their profile records at any time or run the application fully offline in zero-external-transmission mode.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

We identified three primary AI risks and engineered strict technical mitigations:
1. Hallucination in Grant Decisions: LLMs can invent criteria or miscalculate funds.
Mitigation: We established a strict architectural boundary. All quantitative eligibility rules (equity ratios, tenure, turnover caps, co-funding percentages) are calculated 100% deterministically by Python rule engines. AI never decides eligibility.
2. Provider Downtime & Rate Limits: Cloud API outages could stall applications.
Mitigation: We engineered a resilient 3-tier fallback chain: the primary DeepSeek API automatically falls back to our RunPod Serverless GPU vLLM endpoint, which gracefully defaults to deterministic offline templates, guaranteeing zero service interruption.
3. Unchecked Automated Filings: Blind submission risks regulatory penalties.
Mitigation: Enforcing Human-in-the-Loop Supremacy, AI narratives are strictly editable drafts; no claim can be compiled or submitted without explicit human review and cryptographic sign-off.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ☐ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☐ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☐ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☐ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☐ Any use of official SDC sponsor resources has been declared in Section 5.
- ☐ The prototype described here is accessible to judges through the links provided in Section 1.
- ☐ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

Khoo

[Signature]

25/10/2026